

N100 Financial Intelligence Platform

Final Project Report

Name	Onteddu Pullareddy
Project Type	Financial Data Analytics Dashboard
Technology Stack	Python, SQLite, Pandas, Streamlit, Plotly, Git, GitHub
Completion Status	100%

1. Project Overview

The N100 Financial Intelligence Platform is a data analytics and visualization application developed to analyze companies listed in the Nifty 100 index. The platform provides investors and analysts with interactive dashboards, company financial analysis, valuation metrics, sector-wise insights, peer comparison, and downloadable reports.

The project integrates data engineering, analytics, and visualization into one platform using Python and Streamlit.

2. Objectives

The primary objectives of the project are:

- Build a centralized financial analytics platform.
- Store financial datasets in SQLite.
- Analyze company financial ratios.
- Compare companies within sectors.
- Perform valuation analysis.
- Visualize financial trends.
- Generate downloadable reports.
- Create an interactive dashboard for users.

3. Technology Stack

Technology	Purpose
Python	Backend Development
Pandas	Data Processing
SQLite	Database
Streamlit	Dashboard Development
Plotly	Interactive Charts
Git	Version Control
GitHub	Source Code Repository

4. Database

The project uses SQLite as the backend database.

Main Tables

- Companies

- Financial Ratios
- Market Capitalization
- Sector Information
- Balance Sheet
- Profit & Loss
- Cash Flow
- Documents
- Analysis
- Pros & Cons

Total Financial Records

1184+ records

5. Data Processing

The platform performs:

- Data Cleaning
- Missing Value Handling
- Data Validation
- Table Joins
- Feature Engineering
- Financial Ratio Analysis
- Market Cap Integration

6. Dashboard Modules

Home Dashboard

Displays:

- Total Companies
- Average ROE
- Average Debt/Equity
- Net Profit Margin
- Asset Turnover
- Sector Distribution
- KPI Cards

Company Profile

Displays:

- Company Information
- Financial Metrics
- ROE
- ROCE
- Book Value
- Company Overview

Stock Screener

Allows filtering by:

- Sector
- PE Ratio
- PB Ratio
- ROE
- Debt to Equity
- Company Name

Peer Comparison

Compares companies within the same sector using:

- ROE
- PE Ratio
- PB Ratio
- Profit Margin
- Debt to Equity

Financial Trends

Interactive visualization of:

- Revenue Trends
- Profit Trends
- ROE Trends
- Margin Trends

Sector Analysis

Provides:

- Sector Distribution
- Company Count
- Market Cap Category
- Sector-wise Comparison

Capital Allocation

Shows:

- Free Cash Flow
- Capital Expenditure
- Operating Cash Flow
- Debt Analysis

Reports Module

Allows users to:

- Export CSV
- Export Excel
- View Processed Data

Valuation Dashboard

Calculates:

- Free Cash Flow Yield
- Sector Median PE
- Sector Median PB
- Valuation Classification

Valuation Categories: Discount, Fair, Caution, Unknown

7. Valuation Engine

The valuation engine performs:

- PE Ratio Analysis
- PB Ratio Analysis
- Sector Median Calculation
- FCF Yield Calculation
- Dividend Yield Analysis
- Automatic Valuation Classification

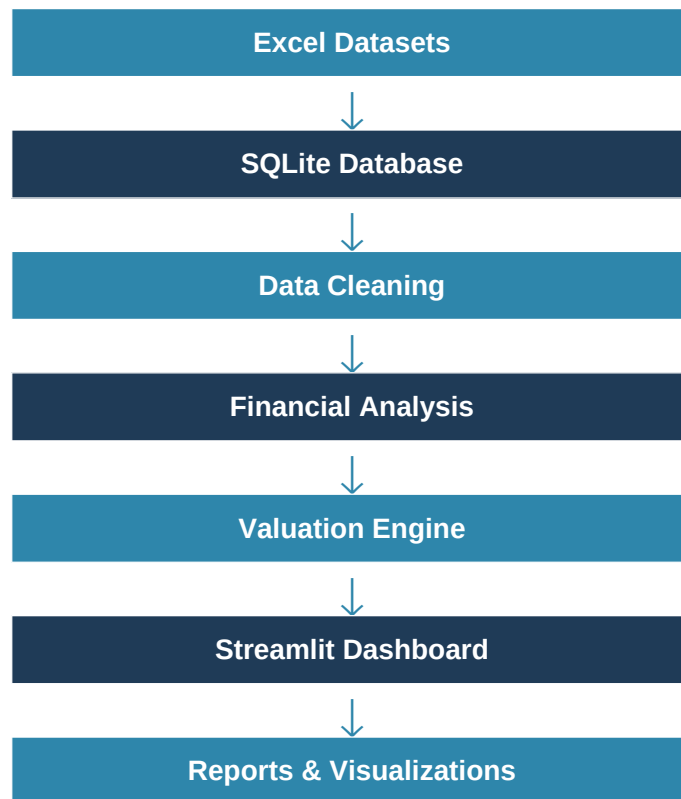
Output Files

- valuation_flags.csv
- valuation_summary.xlsx

8. Features

Feature	Feature
Interactive Dashboard	Peer Comparison
Financial Analytics	Trend Analysis
KPI Cards	Sector Analysis
Pie Charts	CSV Export
Bar Charts	Excel Export
Company Search	Responsive Dashboard
Filters	

9. Project Workflow



10. Challenges Faced

During development, several technical challenges were encountered:

- Database merge issues
- Data type mismatch during joins
- Missing market capitalization records
- Streamlit page navigation configuration
- Duplicate column handling after merges
- Year format inconsistencies
- Dashboard UI customization
- KPI visualization improvements

Each issue was identified, debugged, and resolved through data preprocessing, merge logic refinement, and dashboard enhancements.

11. Project Outcomes

Successfully developed:

- Financial Intelligence Dashboard
- Company Analysis Module
- Stock Screener
- Peer Comparison Module
- Trend Analysis Dashboard
- Sector Analysis
- Capital Allocation Module
- Valuation Engine
- Export Functionality
- Interactive Visualizations

12. Learning Outcomes

This project enhanced practical skills in:

- Python Programming
- Data Analytics
- Financial Data Processing
- SQL & SQLite
- Streamlit Dashboard Development
- Plotly Visualization
- Data Cleaning
- Feature Engineering
- Financial Ratio Analysis
- Git & GitHub Version Control
- Debugging and Error Resolution

13. Future Enhancements

Possible future improvements include:

- Live stock market API integration
- Real-time financial data updates
- Machine learning-based stock prediction
- Portfolio tracking
- User authentication and role-based access
- Cloud deployment
- Mobile-friendly interface
- Interactive financial news integration

14. Conclusion

The N100 Financial Intelligence Platform successfully integrates financial data management, analytics, valuation techniques, and interactive visualization into a single application. The project demonstrates practical application of Python, SQLite, Streamlit, and Plotly to solve real-world financial analysis problems. It provides users with meaningful insights through KPIs, company comparisons, valuation metrics, and downloadable reports, making it a comprehensive financial analytics platform.

Project Summary

Project	N100 Financial Intelligence Platform
Domain	Financial Analytics
Language	Python
Database	SQLite
Framework	Streamlit
Visualization	Plotly
Version Control	Git & GitHub
Analytics Records Processed	1184+
Modules Developed	10+
Project Status	■ Completed (100%)